

B02 MODULE QA REPORT

Python Pipeline Transition | Course 2B DA-to-DE Bridge | INTERNAL QA RC1 | 10 September 2026

Verdict

PASS at module artifact/static/runtime-fixture QA level. Course 2B is not FINAL; learner-PC execution and integrated B01-B06/Mentor QA remain open.

Canonical lessons	10/10; exact five-part analyst-to-engineer transition + 16 teaching stages
Practice / review	10 matched notebook tasks; attempt-first review; fresh changed retries
Fixture runtime	21/21 safe local checks PASS; 96-row batch; duplicate-master and malformed-data traps proven
Gate topology	A normal; B/C sealed; distinct Logistics / E-commerce / Clinic fixtures
PDF set	23p learner; 11p review; 1p Start; 3 x 1p Gate reviews
Sources	Python 3.14.7 / pandas 3.0.5 / Requests 2.34.2 current authorities; external media required = 0
Automated structural QA	312/312 PASS

Fast-track curriculum decision

The 11-lesson baseline is rebuilt into 10 bridge lessons. C1 notebook/pandas familiarity is not retaught as a beginner analyst course. B02 raises the standard to explicit script/function contracts, deterministic multi-file discovery, merge/cardinality validation, file-format/grain handling, bounded API acquisition basics, failure classification, logging/validation gates, safe reruns and a defensible small batch pipeline. Linux/WSL2, Git, venv/dependency pinning, CLI design and Docker remain C3-00; API auth/pagination/rate limits/retries and incremental/state/backfill depth remain C3-01.

Source/video audit

Current authorities were rechecked against Python 3.14.7 documentation, pandas 3.0.5 documentation and Requests 2.34.2 documentation. B02-L06 now uses Real Python "Consuming APIs - GET and POST" 01:50-03:02 as the focused recommended visual for GET -> Response -> JSON/status. The older Corey Requests video is preserved as backup. No external source is required for mastery or a Gate; local book/fixtures remain complete fallback.

Executed fixture evidence

The supplied practice assets were exercised locally: deterministic discovery of 3 monthly files produced 96 unique rows and the protected net-sales control; duplicate customer key C013 is detected before many-to-one merge; malformed numeric and missing-column cases are exposed; nested JSON reconciles 5 orders to 10 item-grain rows; local HTTP fixture retrieval works; all practice/Gate ZIPs pass CRC and all six preserved notebooks parse as nbformat 4.

Render / file result

All learner-facing PDFs were rerendered at 200 DPI after the final source/transition edits. Visual inspection found no known clipping, overlap, broken glyphs or orphan tail pages. Structural QA also checks lesson/page mapping, embedded HTTPS links, practice/review pairing, Gate isolation, fixture byte parity, ZIP CRC, manifest parity, duplicate-content hashes and stale placeholders.

Open blockers before Course 2B release

Target-source versions exceed the safe execution environment, so local runtime evidence does not claim Python 3.14.7/pandas 3.0.5/Requests 2.34.2 execution. Excel writer behavior remains learner-environment dependent. B03-B06, the common Mentor shell, responsive/PWA/backup-restore QA, whole-course capstone/Gate simulation and C3-entry parity are still open.

Promotion rule

B02 is the current internal rebuild candidate only. Any later learner-facing route/content change requires fresh render, manifest/hash and QA evidence.